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Department of Civil and Environmental Engineering (CEE)

Course no: **CEE 4712**

Course Title: **Structural Analysis & Design Sessional II**

ASSIGNMENT NO: 02

Structural Design and Analysis of an 8-Storied Reinforced Concrete Residential Building (BNBC 2020) using ETABS

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Introduction

This project presents the comprehensive structural analysis and design of an 8-story (G+7) reinforced concrete residential building, strictly adhering to the BNBC 2020 standards.

Utilizing ETABS for precise computational modeling, the scope covers the entire design lifecycle—from initial load calculations and rigorous structural analysis to serviceability assessments and final reinforcement detailing.

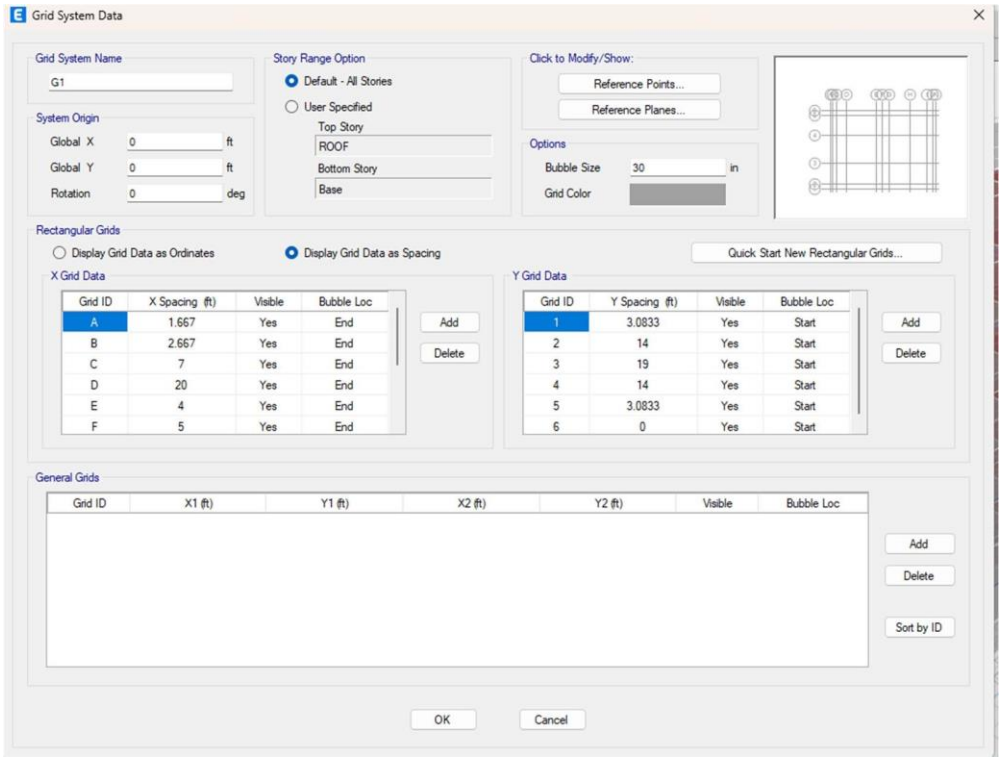
Given the site's location in Dhaka (Seismic Zone 2), the design incorporates a robust analysis of both seismic and wind forces to ensure structural integrity and code compliance.

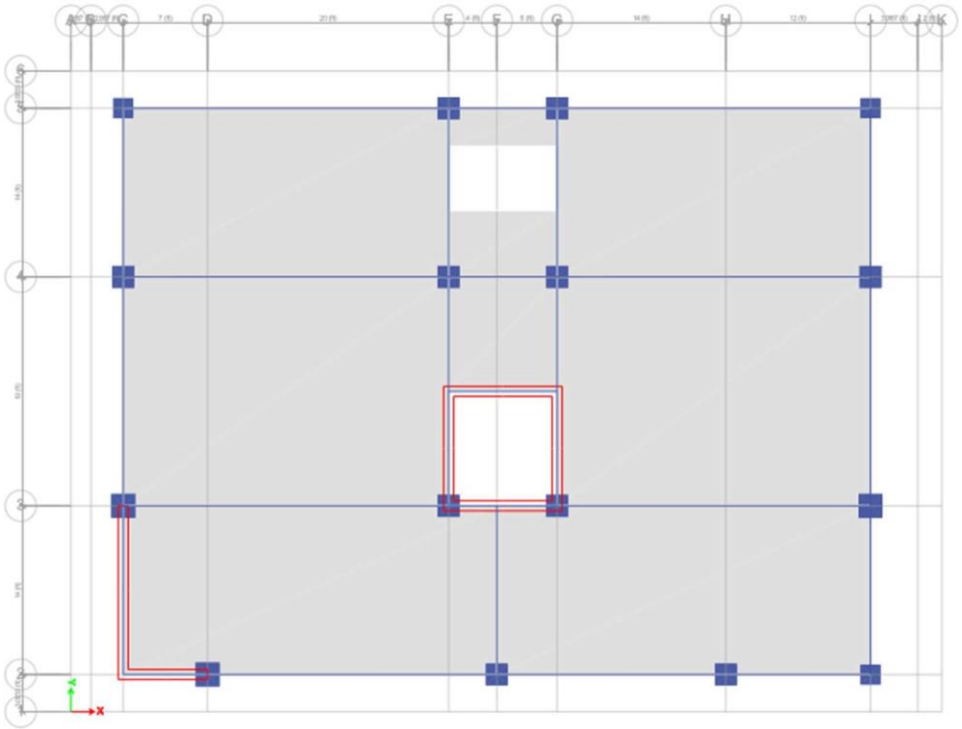
Building Description

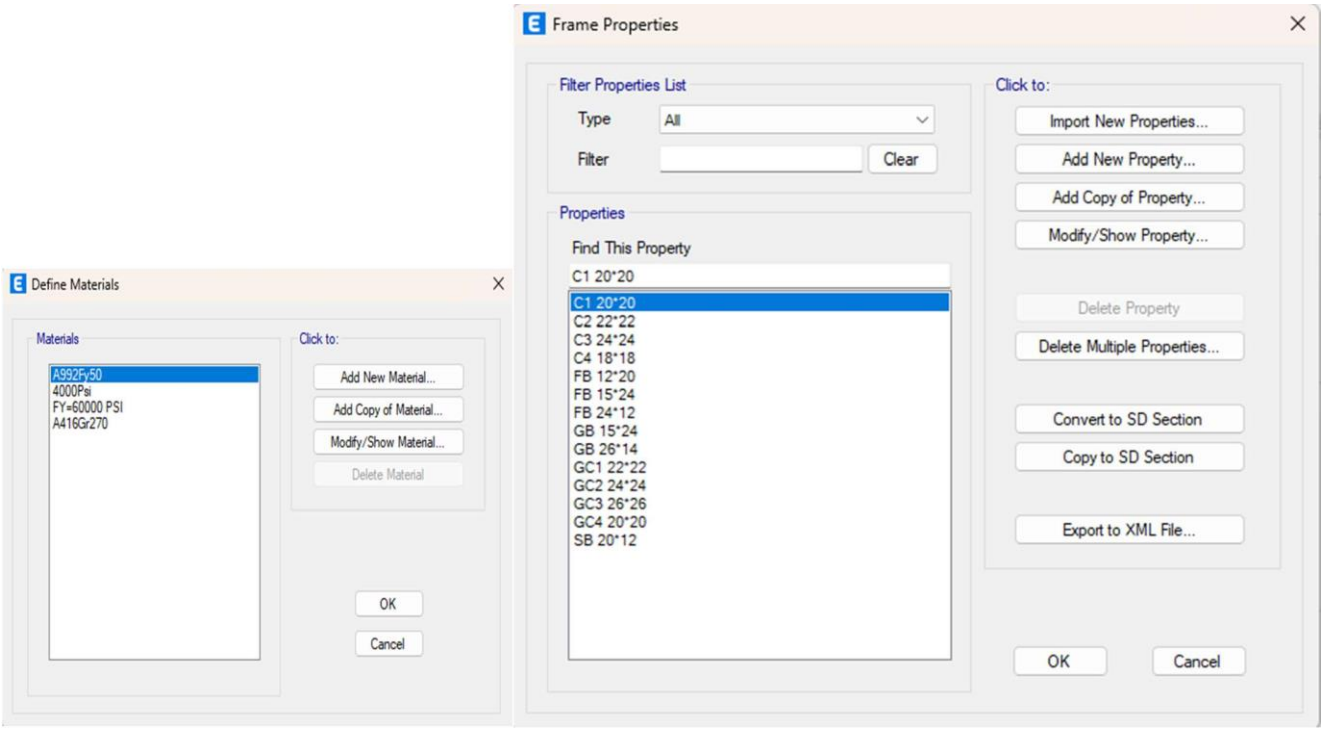
- The proposed building is a G+7 reinforced concrete residential building with a total height of 87 ft (≈26.5 m).
- The structure is designed for a basic wind speed of 147 mph with Exposure Category B.
- A dual structural system consisting of special Moment Resisting Frame (SMRF) and Special Reinforced Concrete Shear Walls is used for lateral load resistance.
- Since the building height is less than 50 m, the selected structural system satisfies the requirements of BNBC 2020.
- The moment-resisting frames carry more than 25% of the gravity loads, meeting the code requirement for dual systems.
- The structural system consists of slab–beam–column framing with shear walls to improve stiffness and control lateral drift.

Modelling Assumptions

Plan Grid Data (X and Y)





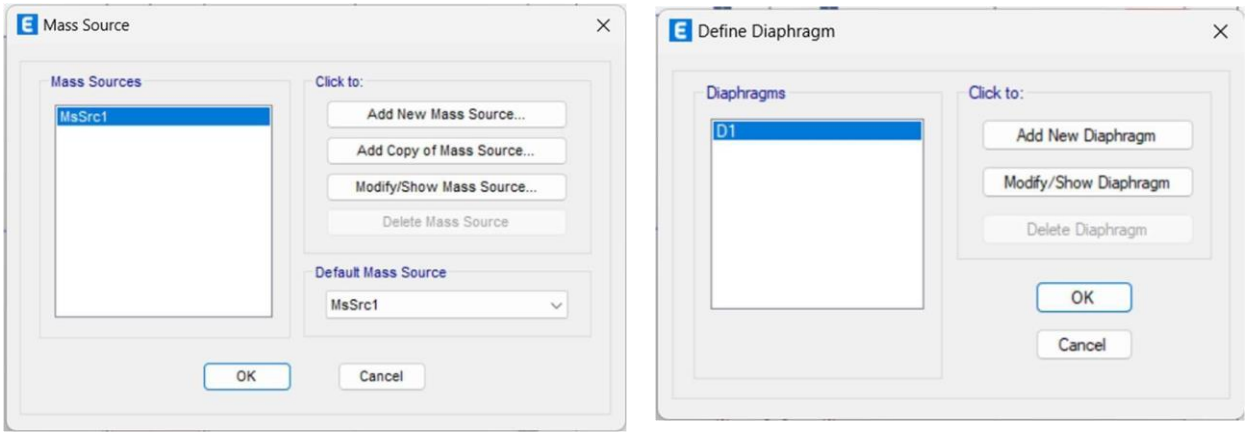


Type of Sections	Material	Strength(psi)	Unit Weight(lb/ft^3)
Columns	Concrete	4000	150
Beams	Concrete	4000	150
Slabs	Concrete	4000	150
Shear Walls	Concrete	4000	150

Slab type: Shell-thin

Support Condition: Fixed at Base

Mass-source and Diaphragm:



Load Calculations (BNBC 2020):

Ground & Typical Floor		
Slab	**Dead Load (psf)**	**Live Load (psf)**
Regular two-way slabs	25	42
Stair slabs	25	100
Roof		
Slab	**Dead Load (psf)**	**Live Load (psf)**
Regular two-way slabs	25	100

Dead Load	(kip/ft)
Exterior Wall	1
Partition Wall	0.5
Parapet Wall	0.15

Wind Load & Earthquake Load

Wind Speed (mph)	147
Importance Factor	1
Topological Factor, Kzt	1
Directionality Factor, Kd	0.85
Gust Factor	0.85
Occupancy Category	I
Site Class	SE
Seismic Zone Co-efficient	0.2
SDC	D
Response Modification Factor, R	8
System Overstrength	3
Deflection Aplification,Cd	5.5
Time Period, T	0.89
Site Coefficient, Fa	1.4
Site Coefficient, Fv	1.75

Dead Load & Live Load Calculation

Dead Load Calculations:

Self-weight of all members (Already Considered by ETABS) Floor

Finish (FF) = 25 psf

Partition Wall

Weight of 5''wall

= Unit weight of brick * thickness of the wall * height of the wall

= (18.9x6.4) lb/ft^3 x (5/12) ft x (10) ft= 504 lb/ft = 0.504 kip/ft

Exterior Wall

Weight of 10''wall

= Unit weight of brick * thickness of the wall * height of the wall

= (18.9x6.4) lb/ft^3 x (10/12) ft x (10) ft= 1008 lb/ft = 1.008 kip/ft

Parapet Wall

= (18.9x6.4) lb/ft^3 x (5/12) ft x (3) ft

=151.2 lb/ft = 0.1512 kip/ft

Partition Wall loads on Exterior Beam

=10'*10"/12*120 pcf=1000lb/ft = 1kip/ft

Partition Wall loads on the Interior Beam

=10'*5"/12*120 pcf= 500lb/ft = 0.5 kip/ft

2.2.4 Weight of Materials and Constructions

In estimating dead loads, the actual weights of materials and constructions shall be used, provided that in the absence of definite information, the weights given in Tables 6.2.1 and 6.2.2 shall be assumed for the purposes of design.

Table 6.2.1: Unit Weight of Basic Materials

Material	Unit Weight (kN/m^3)	Material	Unit Weight (kN/m^3)
Aluminium	27.0	Granite, Basalt	26.4
Asphalt	21.2	Iron - cast	70.7
Brass	83.6	- wrought	75.4
Bronze	87.7	Lead	111.0
Brick	18.9	Limestone	24.5
Cement	14.7	Marble	26.4
Coal, loose	8.8	Sand, dry	15.7

Live Load Calculations:

All others except stairs and balconies

= 2 KN/m^2

= (2 * 20.89) psf

= 41.78 psf

Stairs and exit ways

= 4.80 KN/m^2

= (4.80 * 20.89) psf

= 100.27 psf

Table 6.2.3: Minimum Uniformly Distributed and Concentrated Live Loads[g]

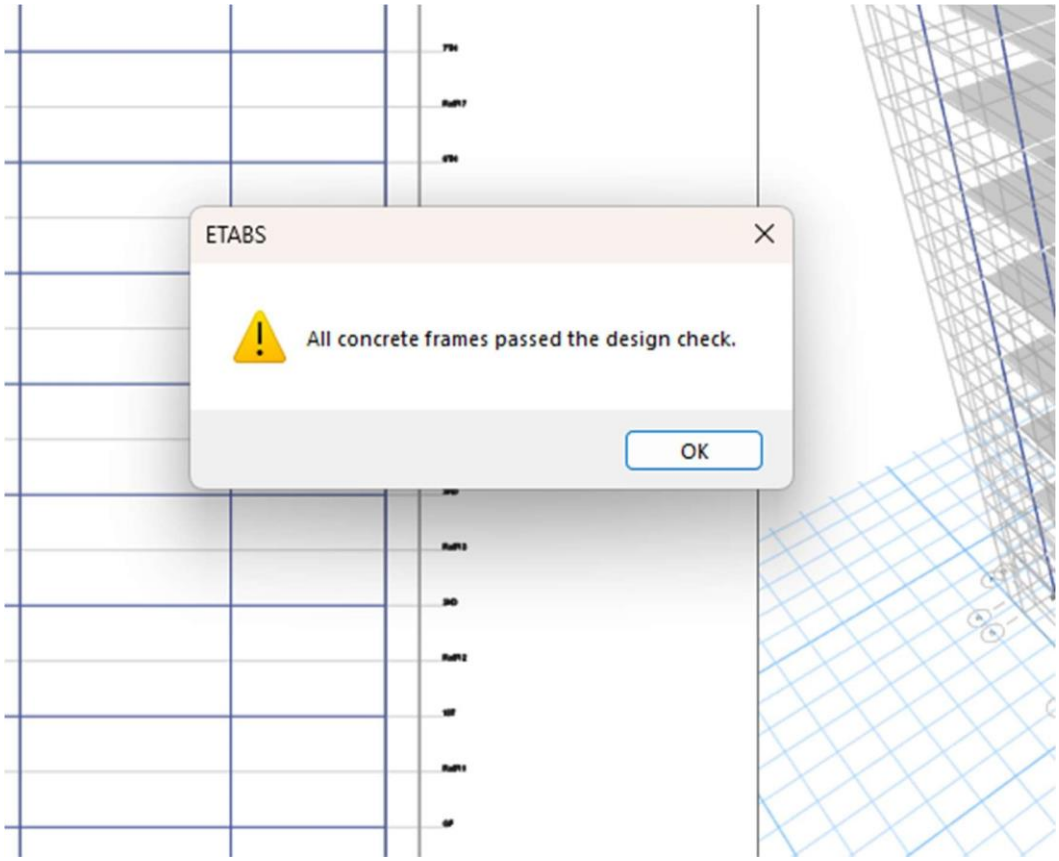
Occupancy or Use	Uniform kN/m²	Concentrated kN
Assembly areas and theaters		
Fixed seats (fastened to floor)	2.90	--
Lobbies	4.80	--
Movable seats	4.80	--
Platforms (assembly)	4.80	--
Stage floors	7.20	--
Balconies (exterior)	4.80	--
On one- and two-family residences only, and not exceeding 19.3 m²	2.90	--
Stairs and exit ways	4.80	See Note[l] below
One- and two-family residences only	2.00	--
Roofs		
Ordinary flat roof	1.00 [h]	--
Pitched and curved roofs	See Table 6.2.4	
Roofs used for promenade purposes	2.90	--
Roofs used for roof gardens or assembly purposes	4.80	--
Roofs used for other special purposes	See Note [i] below	
Residential		
Dwellings (one- and two-family)		
Uninhabitable attics without storage	0.50	--
Uninhabitable attics with storage	1.00	--
Habitable attics and sleeping areas	1.50	--
All other areas except stairs and balconies	2.00	--

Load Combinations:

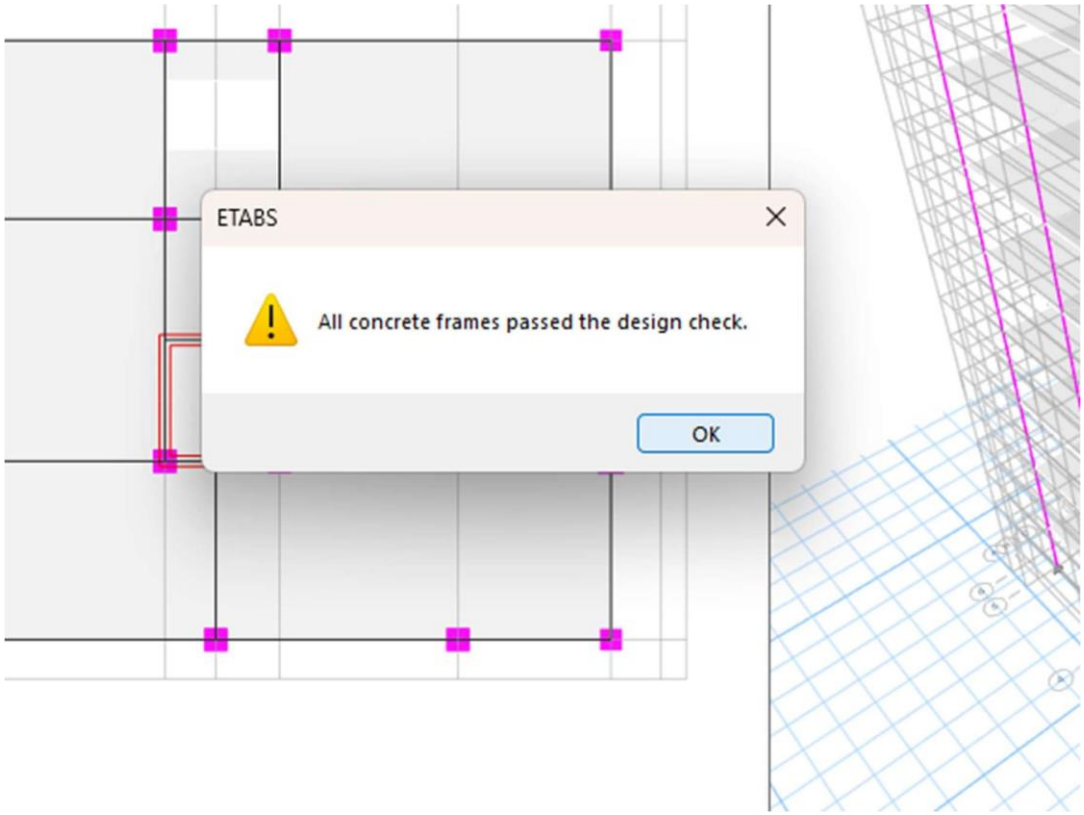
1	1.4D
2	1.2D+1.6LL+0.5Lr
3	1.2D+1.6Lr+L
4	1.2D+0.8W+1.6Lr
5	1.2 D + 1.6 Lr - 0.8W
6	1.2 D + 1.6 W + L + 0.5Lr
7	1.2 D - 1.6 W + L + 0.5 Lr
8	1.246 D + 1.0 Eh +1.0 L
9	1.246 D - 1.0 Eh +1.0 L
10	0.9 D + 1.6 W
11	0.9 D - 1.6 W
12	0.854 D + Eh
13	0.854 D - Eh
14	D + L
15	D + 0.5 L
16	D + 0.5L + 0.7W
17	D + 0.5L - 0.7W

ETABS Analysis Result:

Before Shear Wall-



After Shear Wall-



Layout Modification and Optimization

- Following the preliminary structural modeling in ETABS, a comprehensive performance evaluation was conducted against **BNBC 2020** standards. Initial findings indicated that the building exceeded permissible limits for **storey drift** and **lateral displacement**, necessitating a revision of the structural configuration.

- To enhance lateral stiffness, the structural layout was optimized by introducing additional load-resisting elements and refining the existing system. Subsequent re-analysis of this "Optimized Layout" confirmed that all strength and serviceability criteria were met. This iterative process resulted in a more robust design with significantly improved behavior regarding **torsional stability**, drift control, and overall displacement.

Serviceability Checks (Using Excel)

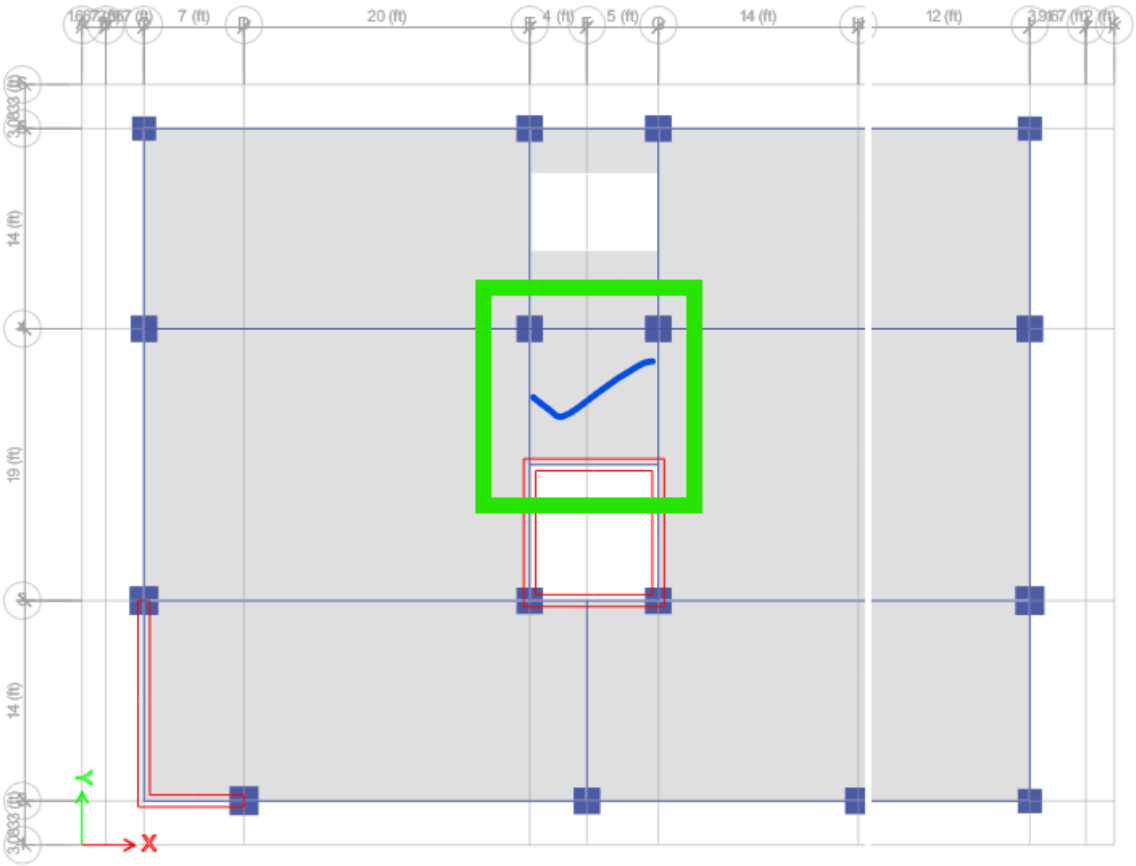
To guarantee the structure's performance under routine operational loads, a comprehensive serviceability assessment was conducted. The primary focus was placed on critical parameters, including **lateral storey displacement, inter-storey drift, and structural deflection limits.**

Data generated from the **ETABS** analytical model was exported to **Microsoft Excel** for rigorous post-processing. These calculated values were then benchmarked against the mandatory thresholds established by **BNBC 2020**. The analysis confirms that the revised structural configuration remains well within the allowable limits, ensuring both occupant comfort and structural integrity. Supporting calculation sheets and comparative data are documented in the **Appendix**.

Manual design checks & Reinforcement detailing

(Beam, Column, Slab, Shear-wall, Stair)

1. Design of Slab



1.1 Type of Slab

For the specified slab,

Shorter span, $L_a = 9\,ft = 108\,in$

Longer span, $L_b = 9\,ft\ 6\,in = 114\,in$

$$\frac{L_b}{L_a} = \frac{114}{108} = 1.05 < 2$$

∴ The slab is designed as a **Two-Way Slab**.

1.2 Assumption of Slab Thickness

Thickness of slab

$h = \frac{2(L_a+L_b)}{180} = \frac{2(108+114)}{180} = 2.467\ in$

But provided slab thickness = **6 in**

1.3 Estimated Moment of Inertia for Beams

For surrounding beams,

Width $b = 12\ in$,

Depth $h = 24\ in$

$b_w + 2h_w = 12 + 2(12) = 36, \quad b_w + 8h_f = 12 + 8(6) = 60,$

$b_w + 2h_w \leq b_w + 8h_f,$

Condition satisfied.

Moment of inertia, $I_b = \frac{bh^3}{12} \times 2$

$I_b = \frac{12\times24^3}{12} \times 2 \qquad So,\ I_b = 27{,}648\ in^4$

1.4 Estimated Moment of Inertia for Slab Sides

$I_{s1} = \frac{114\times6^3}{12}, I_{s1} = 2052\ in^4, \quad I_{s2} = \frac{108\times6^3}{12}, I_{s2} = 1944\ in^4$

1.5 Calculation of Relative Stiffness of Beams

$\alpha_{f1} = \frac{I_b}{I_{s1}},\ \alpha_{f1} = \frac{27648}{2052} = 13.47$

$\alpha_{f2} = \frac{I_b}{I_{s2}},\ \alpha_{f2} = \frac{27648}{1944} = 14.21$

Average stiffness, $\alpha_{fm} = \frac{13.47+14.21}{2} = 13.84 \qquad \alpha_{fm} > 2$

1.6 Check for Slab Thickness

Clear span (long) = **102 in,**

Clear span (short) = **96 in**

$$\beta = \frac{102}{96} = 1.0625$$

Since $\alpha_{fm} > 2$

$h_{min} = \frac{l_n\left(0.8+\frac{fy}{200000}\right)}{36+9\beta}$, $h_{min} = \frac{102\left(0.8+\frac{60000}{200000}\right)}{36+9(1.0625)}$

$$h_{min} = 2.46\,in$$

Provided thickness $6\,in > 2.46\,in$

∴ **Slab thickness is OK**

1.7 Calculation of Design Moments

Moment coefficients from chart:

Coefficient	Value
Ca(D)+	0.029
Ca(L)+	0.047
Ca-	0.072
Cb(D)+	0.008
Cb(L)+	0.012
Cb-	0.02

Factored Load

Dead load, $DL = 150 \times \frac{6}{12} + 25 = 75 + 25 = 100\,psf$

Live load, $LL = 41.77\,psf$

Factored load

$W = 1.4DL + 1.7LL\; W = 1.4(100) + 1.7(41.77) \qquad W = 140 + 71 = 211\ psf$

1.8 Design Moments

Positive moment in short direction

$M_a^+ = Ca(D) + DLL_a^2 + Ca(L) + LLL_a^2\; M_a^+ = 0.029(100)(9^2) + 0.047(42)(9^2)$

$$M_a^+ = 394\,lb-ft$$

Positive moment in long direction

$M_b^+ = Cb(D) + DLL_b^2 + Cb(L) + LLL_b^2\; M_b^+ = 0.008(100)(12^2) + 0.012(41.77)(12^2)$

$$M_b^+ = 187.4\,lb-ft$$

Negative moments

$M_a^- = Ca^- WL_a^2, \quad M_a^- = 0.072(211)(9^2)$,

$M_a^- = 1231\, lb - ft$

$M_b^- = Cb^- WL_b^2$, $M_b^- = 0.02(211)(12^2)$

$M_b^- = 608\, lb - ft$

1.9 Calculation of Main Reinforcement & Spacing

As per convention, use #3 bar as the main reinforcement for the slab.

1.9.1 Short Direction (a)

da = 6 − 1 = 5 inch

Positive Reinforcement, $A_s = \frac{0.85 f_c' bd}{f_y}\left[1 - \sqrt{1 - \frac{2M_u}{0.85\phi f_c' bd^2}}\right]$ = 0.055 in^2

Spacing:

$s = \frac{12 A_{bar}}{A_s} = \frac{12 \text{ x } 0.11}{0.055}$ = 23 inch, but Maximum Spacing, Smax = 2t = 12” c/c

Negative Reinforcement, $A_s = \frac{0.85 f_c' bd}{f_y}\left[1 - \sqrt{1 - \frac{2M_u}{0.85\phi f_c' bd^2}}\right]$ = 0.128 in^2

Spacing:

$s = \frac{12 A_{bar}}{A_s} = \frac{12 \text{ x } 0.11}{0.128}$ = 10.31 inch = 10” c/c

1.9.2 Long Direction (b)

db = 6 – 1.5 = 4.5 inch

Positive Reinforcement, $A_s = \frac{0.85 f_c' bd}{f_y}\left[1 - \sqrt{1 - \frac{2M_u}{0.85\phi f_c' bd^2}}\right]$ = 0.03491 in^2

Spacing:

$s = \frac{12 A_{bar}}{A_s} = \frac{12 \text{ x } 0.11}{0.03491}$ = 37.811 inch, but Maximum Spacing, Smax = 2t = 12” c/c

Negative Reinforcement, $A_s = \frac{0.85 f_c' bd}{f_y}\left[1 - \sqrt{1 - \frac{2M_u}{0.85\phi f_c' bd^2}}\right]$ = 0.074 in^2

$s = \frac{12 A_{bar}}{A_s} = \frac{12 \text{ x } 0.11}{0.074}$ = 17.8 inch

but Maximum Spacing, Smax = 2t = 12” c/c

Providing #3 @10 inch c/c for both directions.

1.10 Calculation of Temperature and Shrinkage Reinforcement:

Ast = 0.0018bt = 0.0018 × 12 × 6 = 0.1296 in

Spacing, s =12 × 0.11/0.1296 = 10.185 ≈ 10 in

Providing #3 @10 inch c/c temperature and shrinkage reinforcement.

1.11 Effective Depth Check:

$\rho_{max} = 0.75\rho_b = 0.016$

$d^2 = \frac{M_u}{\phi f_y \rho b\left(1-\rho\frac{\beta}{\alpha}\frac{f_y}{f_c'}\right)}$ Here, $\alpha = 0.85,\ \beta = 0.425,\ \frac{f_y}{f_c'} = \frac{60}{4}$

$\therefore d_{min} = 1.096 \times 10^{-4}$ Mu

1.11.1 Short direction (a)

$\therefore d_{min}$ = 3.89 inch < 5 inch (OK)

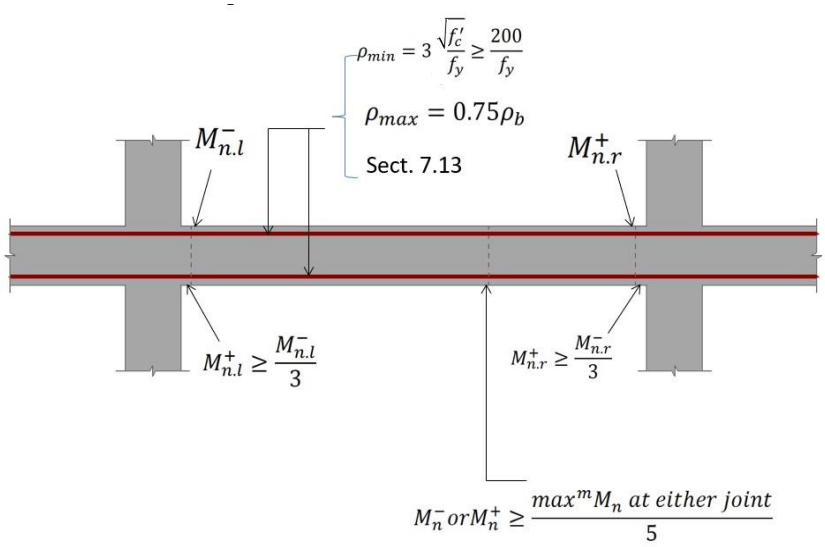
1.11.2 Long direction (b)

$\therefore d_{min}$ = 2.11 inch < 4.5 inch (OK)

Final Reinforcement Summary

Item	Reinforcement
Slab thickness	6 in
Main reinforcement (short direction)	#3 @ 10 in c/c
Main reinforcement (long direction)	#3 @ 10 in c/c
Temperature & shrinkage reinforcement	#3 @ 10 in c/c

2. Design of Beam



2.1 Beam Details

Beam section

$b = 12\,in, h = 24\,in$

Beam length $L = 19\,ft$

Concrete strength $f'_c = 4000\,psi$

Steel strength $f_y = 60000\,psi$

Clear cover $= 1.5\,in$

Distance to reinforcement centroid $d \approx 22\,in$

Design code: **BNBC 2020 / ACI 318**

2.2 Design Moments from ETABS

From ETABS beam design output:

Negative moment (support)

$$M_u = 183.69 \; kip - ft$$

Positive moment (midspan)

$$M_u = 70.74 \; kip - ft$$

Required steel areas from ETABS:

Top reinforcement required

$$A_s = 1.93\, in^2$$

Bottom reinforcement required

$$A_s = 1.297\ in^2$$

Minimum reinforcement

$$A_{s,min} = 0.90\, in^2$$

∴ Bottom reinforcement **controlled by minimum steel**.

2.3 Flexural Design at Support (Negative Moment)

Required steel area $A_s = 1.93\ in^2$

Use #8 bars

Area of #8 bar

$A_{bar} = 0.79\, in^2$

Number of bars $n = \frac{1.93}{0.79} = 2.443$

Provide $3\#8\ bars$

Provided steel area

$A_s = 3(0.79)\,, A_s = 2.37\, in^2$

Provided steel ≥ required steel

2.4 Flexural Design at Midspan (Positive Moment)

Required steel $A_s = 0.90\, in^2$

Use #8 bars

$n = \frac{0.90}{0.79}\, n = 1.14$

Provide $2\#8\ bars$

Provided steel

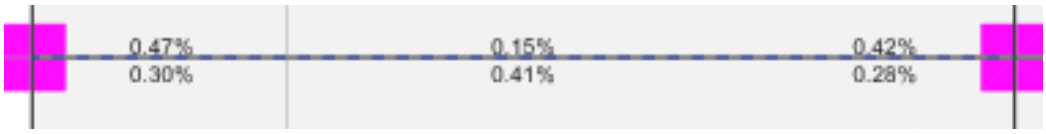
$A_s = 2(0.79)\ A_s = 1.58\ in^2$

Provided steel ≥ required steel

2.5 Reinforcement Ratio Check

$$\rho = \frac{A_s}{bd}$$

For support reinforcement, $\rho = \frac{2.37}{12 \times 22} = 0.009$



2.6 Shear Design

From ETABS

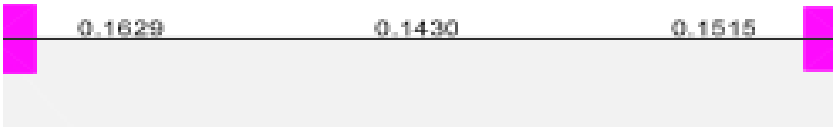
Factored shear. $V_u = 39.983\ kip$

Concrete shear strength, $\phi V_c = 25.614\ kip$

Required shear from stirrups, $\phi V_s = 10.125\ kip$

Shear reinforcement required, $\frac{A_v}{s} = 0.4739\ in^2/ft$

No torsion reinforcement required.



2.7 Stirrup Design

Use #3 stirrups

Area of #3 bar

$A_{bar} = 0.11\ in^2$

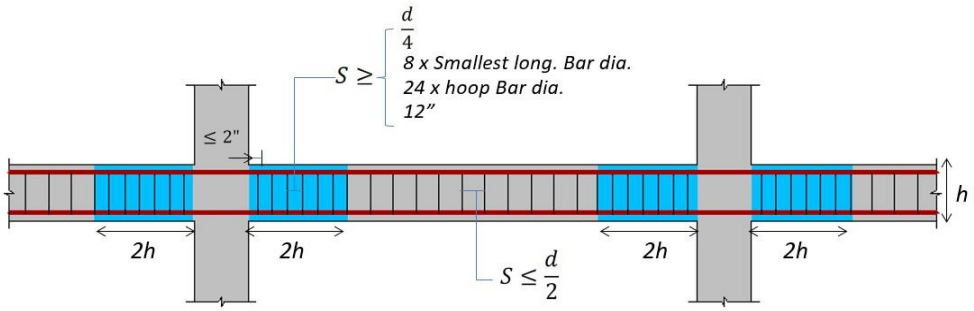
Two-legged stirrup area

$A_v = 2(0.11)\ A_v = 0.22\ in^2$

Spacing, $s = \frac{A_v}{A_v/s}\ s = \frac{0.22}{0.4739}\ s = 0.464\ ft\ s = 5.57\ in$

Provide, #3@6 *in*

2.8 Seismic Stirrup Spacing



Near supports

$s = d/4\ s = 22/4\ s \approx 5.5\ in$

Provide #3@6 *in*

within 2d distance from supports

Middle region

Provide #3@8 *in*

Final Beam Reinforcement Summary

Item	Reinforcement
Beam section	12 in × 24 in
Beam span	19 ft
Top reinforcement (support)	3 #8 bars
Bottom reinforcement (midspan)	2 #8 bars
Stirrups near support	#3 @ 6 in
Stirrups midspan	#3 @ 8 in

3. Design of Column

3.1 Column Details

Column section:

$b = 24\,in \quad h = 24\,in$

Column height: $L = 120\,in = 10\,ft$

Column type: Sway special column

Clear cover$= 2\,in$

Distance to reinforcement centroid, $d_c = 2.5688\,in$

3.2 Material Properties

Concrete compressive strength, $f'_c = 4000\,psi$

Steel yield strength, $f_y = 60000\,psi$

Strength reduction factor, $\phi = 0.65$

3.3 Design Forces from ETABS

Factored axial load, $P_u = 395.567\,kip$

Moment about the minor axis, $M_{u2} = -43.51\,kip - ft$

Moment about the major axis, $M_{u3} = 135.76\,kip - ft$

Minimum design moments, $M_{min} = 43.51\,kip - ft$

Required steel area, $A_s = 5.76\ in^2$

Steel ratio, $\rho = 1.25\%$

BNBC limit, $1\% < \rho < 8\%$ {OK}

3.4 Selection of Longitudinal Reinforcement

Use #10 bars

Area of #10 bar, $A_{bar} = 1.27\,in^2$

Number of bars required

$n = \frac{A_s}{A_{bar}}\ n = \frac{5.76}{1.27}\ n = 4.53$

Provide 8#10 *bars*

Provided steel area,

$A_s = 8(1.27) = 10.16\, in^2$

Steel ratio,

$\rho = \frac{10.16}{24\times24} = 1.76\%$ [**Within allowable limits**]

3.5 Nominal Axial Strength

Gross area, $A_g = 24 \times 24 = 576\, in^2$

Nominal axial capacity

$P_n = 0.85f'_c(A_g - A_s) + f_y A_s\ P_n$

$= 0.85(4000)(576 - 10.16) + 60000(10.16)\ P_n = 0.85(4000)(565.84) + 609600$

$P_n = 1923856 + 609{,}600 \quad P_n = 2533456\, lb = 2533.456\, kip$

Design axial capacity

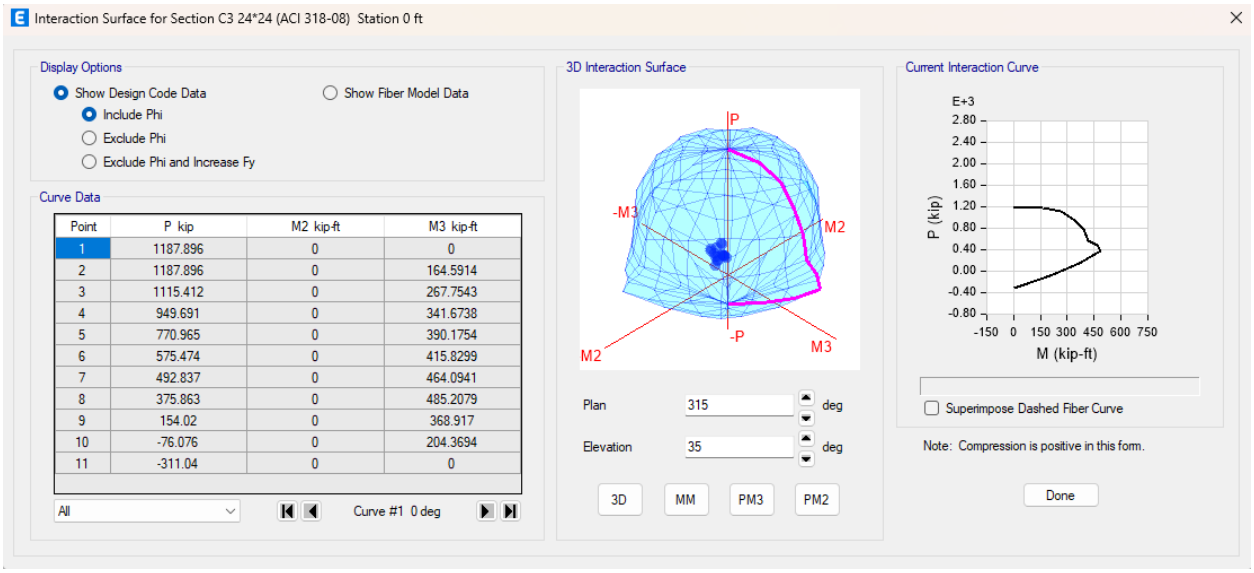
$\phi P_n = 0.65(2533.456)\ \ \phi P_n = 1646.74\, kip$

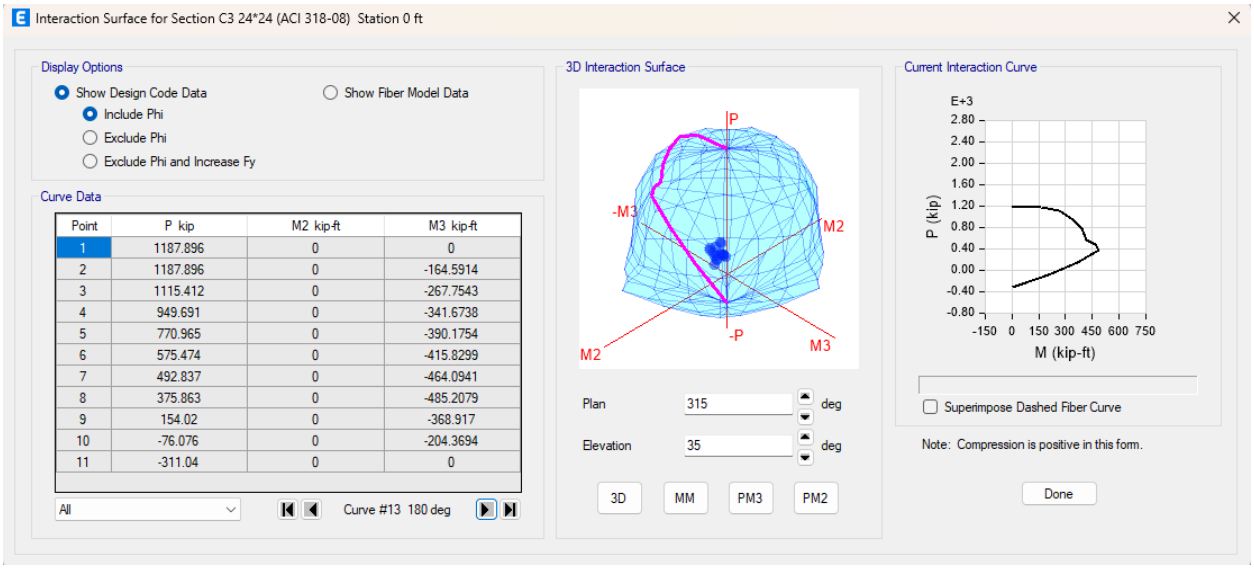
Check,

$\phi P_n > P_u = 1646.74 > 395.567$

[**Column safe in axial compression**]

3.6 Biaxial Moment Check using Interaction Diagram





From the interaction diagrams provided:

At $P \approx 400\,kip$,

Moment capacity from the curve, $M_n \approx 300–350\,kip-ft$

Applied moment, $M_{u3} = 135.76\,kip-ft$

Since, $M_u < M_n$

The design point lies inside the interaction surface.

[**Column safe for biaxial bending**]

3.7 Bresler Interaction Equation

$$\frac{M_{u2}}{\phi M_{n2}} + \frac{M_{u3}}{\phi M_{n3}} \leq 1$$

Approximate moment capacities from the diagram

$\phi M_{n2} \approx 300\,kip-ft$ $\phi M_{n3} \approx 300\,kip-ft$ $\frac{43.51}{300} + \frac{135.76}{300} = 0.145 + 0.452 = 0.597 < 1$

[**Interaction equation satisfied**]

3.8 Shear Check

Major direction, $V_u = 29.861\,kip$

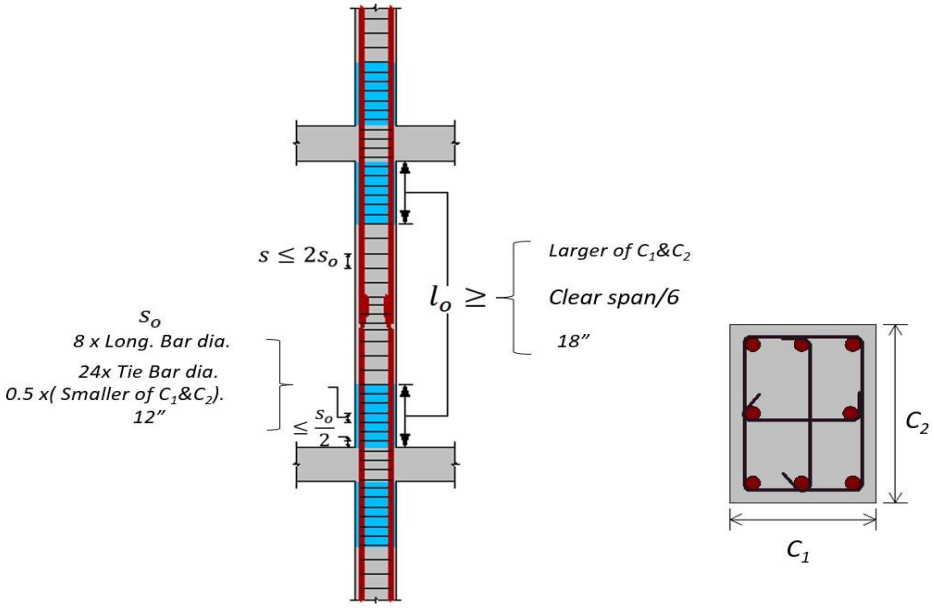
Minor direction, $V_u = 27.724\,kip$

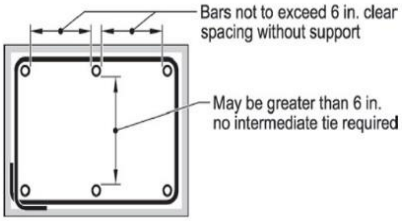
Concrete shear capacity

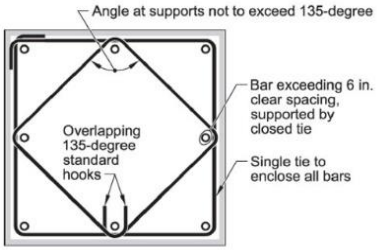
$\phi V_c = 65.55\, kip\ V_u < \phi V_c$

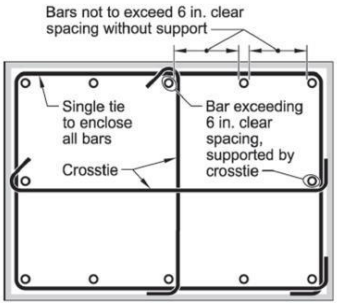
[No additional shear reinforcement required beyond ties.]

3.9 Transverse Reinforcement (Tie Design)









Use #4 ties,

Area, $A_v = 0.20\,in^2$

Maximum tie spacing (ACI)

$s \leq 16d_b\ s \leq 48d_{tie}\ s \leq b$

Controlling value, $s = 12\,in$

Provide #4 ties @ 10 in

At confinement zones (top and bottom),

provide #4 ties @ 5 inches

Final Column Reinforcement Summary

Item	Reinforcement
Column size	24 in × 24 in
Longitudinal bars	8 #10 bars
Steel ratio	1.76 %
Tie reinforcement	#4 ties
Tie spacing (ends)	#4 @ 5 in
Tie spacing (middle)	#4 @ 10 in

4. STAIR Design

Type: **Two-legged stair**

Horizontal length of one flight, $L = 7\,ft$

Width of stair, $B = 9\,ft$

Landing slab size

Length = 3.5 ft
Width = 9 ft

Riser = 7 in
Tread = 10 in

Waist slab thickness, $t = 10\,in$

Inclined Length of Stair

Slope of stair

$\tan\theta = \frac{riser}{tread}$ $\tan\theta = \frac{7}{10}$ $\theta \approx 35°$ $\cos\theta = 0.819$

Inclined span

$L_i = \frac{L}{\cos\theta}$ $L_i = \frac{7}{0.819}$ $L_i = 8.54\ ft$

Load Calculation

(1) Waist Slab Weight

Unit weight of concrete $150\,pcf$

$w_1 = \frac{150\times10}{12}$ $w_1 = 125\,psf$

(2) Step Weight

Volume of a step triangle, $\frac{riser\times tread}{2}$, $w_2 = \frac{150(7\times10)}{2\times12\times12} = 36.5\,psf$

(3) Floor Finish: $w_3 = 25\,psf$

(4) Live Load: $LL = 42\,psf$

Total Dead Load

$DL = w_1 + w_2 + w_3\ DL = 125 + 36.5 + 25\ DL = 186.5\,psf$

Factored Load

$w_u = 1.4DL + 1.7LL\ w_u = 1.4(186.5) + 1.7(42)\ w_u = 261.1 + 71.4\ w_u = 332.5\,psf$

Load per ft width, $w = 332.5\,lb/ft$

Bending Moment

For a simply supported stair slab

$M_u = \frac{wL^2}{8}, M_u = \frac{332.5(8.54)^2}{8}, M_u = 3031.2\,lb-ft, M_u = 36{,}374.63\,lb-in$

Effective Depth

Overall thickness $h = 10\,in$

Clear cover $1.5\,in$

Assume #4 bar

$d = 10 - 1.5 - 0.25\ d = 8.25\,in$

Steel Area Calculation: $\phi = 0.9$

Concrete strength, $f'_c = 4000\,psi$

Steel strength, $f_y = 60000\,psi$

Required steel area, $A_s \approx 0.12\,in^2$

Minimum Reinforcement Check

$A_{s,min} = 0.0018bt\ A_{s,min} = 0.0018(12)(10)\ A_{s,min} = 0.216\,in^2$

Minimum steel controls.

Provide Reinforcement

Use #4 bars

Bar area, $A_{bar} = 0.20\,in^2$

Spacing, $s = \frac{12A_{bar}}{A_s}\ s = \frac{12(0.20)}{0.216}\ \ s \approx 11\,in$

Provide #4 @ 10 in c/c

Distribution Reinforcement

Minimum steel governs.

Provide #3 @ 8 in c/c

Design of Landing Slab

Landing slab size

Length = 3.5 ft
Width = 9 ft

$$\frac{9}{3.5} = 2.57 > 2$$

Slab Thickness

Provided thickness, $h = 8\,in$

Load Calculation

Dead load, $DL = 150 \times \frac{8}{12} + 25 \;\; DL = 100 + 25 \;\; DL = 125\,psf$

Live load, $LL = 42\,psf$

Factored load

$W = 1.4DL + 1.7LL \; W = 1.4(125) + 1.7(42) \; W = 175 + 71.4 \; W = 246.4\,psf$

Design Moment

Using slab coefficient, $M_u \approx 400 \;\; lb - ft$

Minimum steel governs.

Reinforcement

Minimum steel

$A_s = 0.0018bt \; A_s = 0.0018(12)(8) \; A_s = 0.173\,in^2$

Using #3 bars

$A_{bar} = 0.11\,in^2$

Spacing, $s = \frac{12(0.11)}{0.173} \; s \approx 7.6\,in$

Provide #3 @ 7 in c/c

Final Reinforcement Summary

Stair Flight

Item	Reinforcement
Waist slab thickness	10 in
Main reinforcement (along slope)	#4 @ 10 in c/c
Distribution reinforcement	#3 @ 8 in c/c

Landing Slab

Item	Reinforcement
Landing slab thickness	8 in
Main reinforcement	#3 @ 7 in c/c
Distribution reinforcement	#3 @ 7 in c/c

5. DESIGN OF RECTANGULAR SHEAR WALL

1. Introduction

The wall is designed according to BNBC / ACI 318 provisions.

The shear wall behaves as a vertical cantilever fixed at the base, resisting lateral seismic forces.

2. Given Data

2.1 Material Properties

Concrete compressive strength, $f'_c = 4000\ psi$

Steel yield strength, $f_y = 60{,}000\ psi$

Strength reduction factor for shear, $\phi = 0.75$

2.2 Geometry of Shear Wall

Wall length, $l_w = 9ft = 108in$

Wall thickness, $t = 10\ in$

Wall area, $A_{cv} = t \times l_w\ A_{cv} = 10 \times 108 = 1080\ in^2$

2.3 Governing Design Forces (From ETABS)

Factored shear force, $V_u = 275\ kip$

Factored overturning moment, $M_u = 21{,}000\ kip - ft$

3. Effective Length of Shear Wall

For a rectangular shear wall, the effective length equals the actual wall length.

$l_e = l_w\ l_e = 9ft$

This effective length is used for both shear and flexural design.

4. Shear Strength Check

4.1 Concrete Contribution to Shear Strength

According to ACI 318 / BNBC

$V_c = 2\sqrt{f'_c}A_{cv}\ \sqrt{4000} = 63.24\ \ V_c = 2 \times 63.24 \times 1080\ \ V_c = 136598.4\ lb = 136.5\ kip$

Design Shear Strength

$\phi V_c = 136.5 \times 0.75 \qquad \phi V_c = 102.375\ kip$

Comparison

$V_u = 275\ kip \qquad \phi V_c = 102.375\ kip$

Since, $V_u > \phi V_c$

Concrete alone is not sufficient.

Therefore, shear reinforcement is required.

5. Required Shear Reinforcement

Required nominal shear strength from steel:

$V_s = \frac{V_u}{\phi} - V_c, V_s = \frac{275}{0.75} - 136.5 \ , V_s = 230.16\ kip$

Thus shear reinforcement must resist:

$$V_s = 230.16\ kip$$

6. Horizontal Reinforcement Design

Minimum horizontal reinforcement ratio (ACI): $\rho_h = 0.0025$

Required steel area per inch:

$A_s = \rho_h \times t \qquad A_s = 0.0025 \times 10 = 0.025\ in^2/in$

Provide Reinforcement

Bar size: #4 bar

Area of #4 bar, $A_b = 0.20\ in^2$

Spacing: $s = 6\ in$

Steel area per inch:

$A_s = \frac{0.20}{6} A_s = 0.033\ in^2/in$

Since, $0.033 > 0.025$

[**Horizontal reinforcement satisfies requirement**]

7. Flexural Design (Overturning Moment)

Convert moment to kip-inch

$M_u = 21{,}000 \times 12\ M_u = 252{,}000\ kip - in$

The wall behaves as a cantilever column section.

Extreme fiber compression stress:

$$f = \frac{M}{S}$$

Section modulus

$S = \frac{tl_w^2}{6}\ S = \frac{10 \times 108^2}{6}\quad S = 19{,}440\ in^3$

Stress: $f = \frac{252000}{19440}\quad f = 12.96\ ksi$

Since compression stress is high near the ends, boundary elements are required.

8. Boundary Element Requirement

Boundary elements are required when:

• High compressive stress occurs at wall ends
• Large overturning moment exists

9. Boundary Element Dimensions

ACI recommends a boundary width of approximately:

$$0.15l_w \le b_e \le 0.20l_w$$

Take, $b_e = 0.15 \times 9\ b_e = 1.35\ ft$

Adopt, $b_e = 1.5\ ft$

Thus boundary element width: $1.5\ ft\ (18in)$ provided on each end.

10. Boundary Element Reinforcement

Provide:

Longitudinal reinforcement: 6 – #8 bars

Area of #8 bar, $A_b = 0.79\ in^2$

Total steel: $A_s = 6 \times 0.79\ A_s = 4.74\ in^2$

Confinement Reinforcement

Provide #3 closed ties @ 4 in c/c

This ensures ductile confinement during seismic loading.

11. Vertical Reinforcement in Web Region

Minimum vertical reinforcement ratio, $\rho_v = 0.0025$

Provide: #5 bars @ 8 in c/c

Area of #5 bar, $A_b = 0.31\ in^2$

Steel per inch: $A_s = \frac{0.31}{8}\ A_s = 0.038\ in^2/in$

Since $0.038 > 0.025$

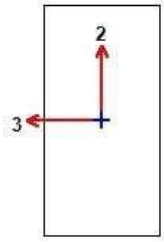
[**Vertical reinforcement satisfies BNBC requirement**]

12. Final Design Summary

Item	Value
Concrete strength	4000 psi
Steel grade	60 ksi
Wall length	9 ft
Wall thickness	10 in
Factored shear V_u	275 kip
Factored moment M_u	21,000 kip-ft
Horizontal reinforcement	#4 @ 6 in
Vertical reinforcement	#5 @ 8 in
Boundary element width	1.5 ft each end
Boundary reinforcement	6-#8 bars
Confinement ties	#3 @ 4 in

ETABS Concrete Frame Design

ACI 318-08 Beam Section Design (Summary)



Beam Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (in)	LLRF	Type
3RD	B26	199	FB 24*12	1.246D-1E+1L	12	228	1	Sway Special

Section Properties

b (in)	h (in)	b_f (in)	d_s (in)	d_{ct} (in)	d_{cb} (in)
12	24	12	0	1.5	1.5

Material Properties

E_c (lb/in²)	f'_c (lb/in²)	Lt.Wt Factor (Unitless)	f_y (lb/in²)	f_{ys} (lb/in²)
3604997	4000	1	60000	60000

Design Code Parameters

ϕ_T	ϕ_{CTied}	$\phi_{CSpiral}$	ϕ_{Vns}	ϕ_{Vs}	ϕ_{Vjoint}
0.9	0.65	0.75	0.75	0.6	0.85

Design Moment and Flexural Reinforcement for Moment, M_{u3}

	Design Moment kip-ft	Design P_u kip	-Moment Rebar in²	+Moment Rebar in²	Minimum Rebar in²	Required Rebar in²
Top (+2 Axis)	-183.6962	0	1.9369	0	0.9	1.9369
Bottom (-2 Axis)	91.8481	0	0	0.9358	0.9	0.9358

Shear Force and Reinforcement for Shear, V_{u2}

Shear V_{u2} kip	Shear ϕV_c kip	Shear ϕV_s kip	Shear V_p kip	Rebar A_v/s in²/ft
39.983	25.614	10.125	16.946	0.4739

Torsion Force and Torsion Reinforcement for Torsion, T_u

T_u kip-ft	$\phi T_{cr}/4$ kip-ft	Area A_o in²	Perimeter, p_h in	Rebar A_t/s in²/ft	Rebar A_l in²
0.0891	4.5537	148.11	58	0	0

ACI 318-08 Concrete Beam Design

Geometric Properties (Part 1 of 2)

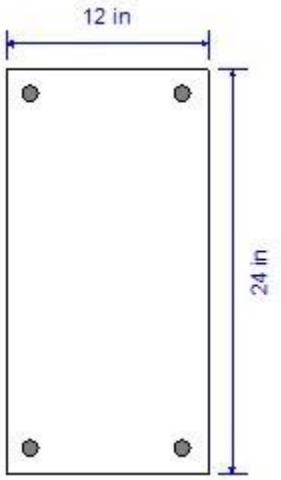
Beam Label	Section Property	Length	Section Width	Section Depth	Distance to Top Rebar Center
199	FB 24*12	19 ft	12 in	24 in	1.5 in

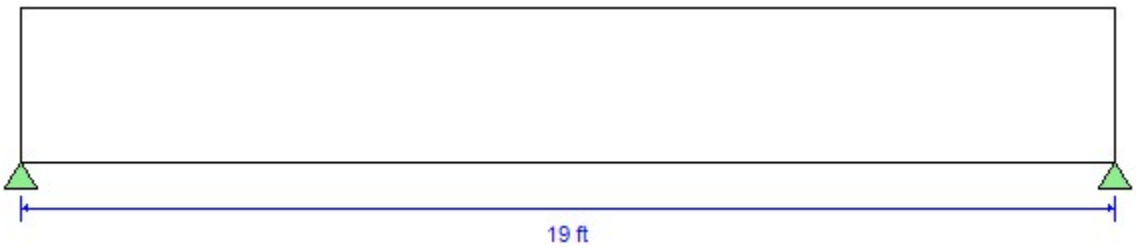
Geometric Properties (Part 2 of 2)

Distance to Bot Rebar Center
1.5 in

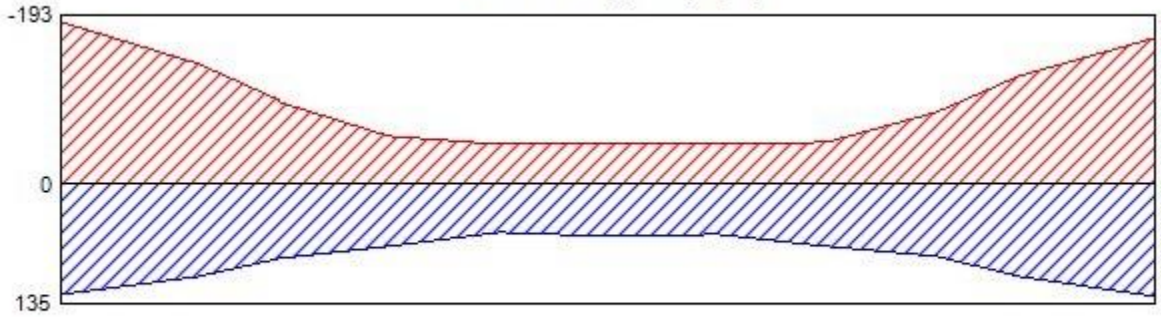
Material Properties

Concrete Comp. Strength	Concrete Modulus	Longitudinal Rebar Yield	Shear Rebar Yield
4000 lb/in^2	3604997 lb/in^2	60000 lb/in^2	60000 lb/in^2



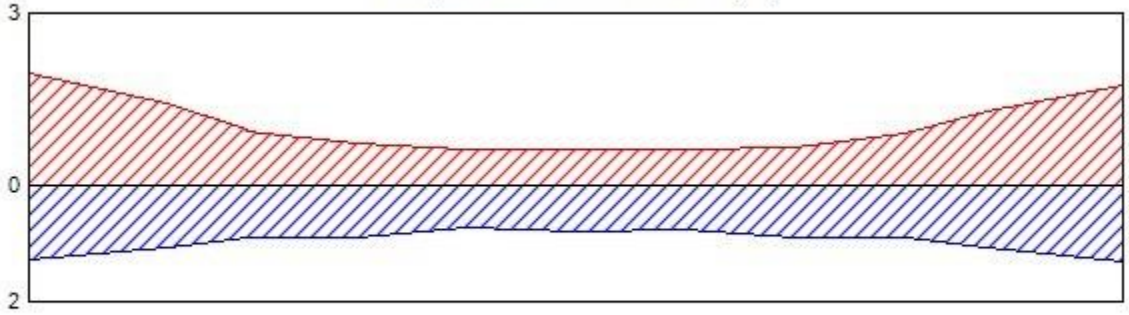


Moment Diagram (kip-ft)



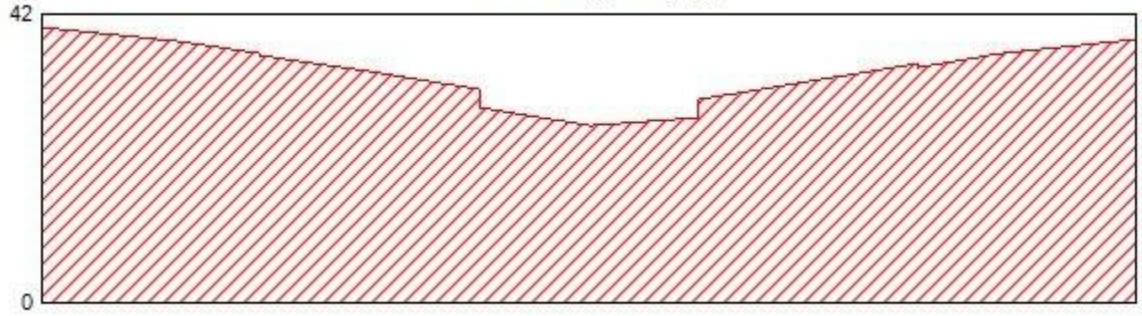
(-)Moment	-183.6962	-55.2706	-167.2676
(+)Moment	125.7549	70.7432	127.7766

Longitudinal Reinforcement (in²)

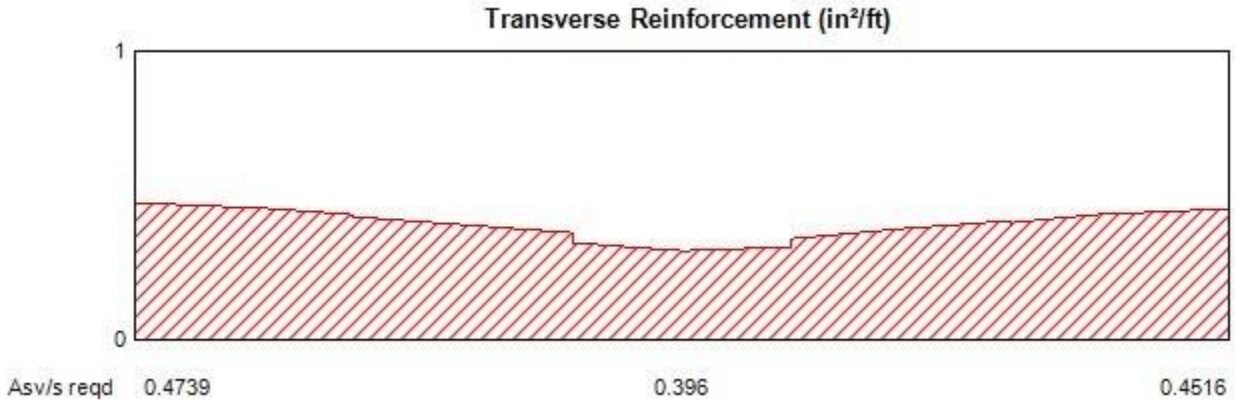


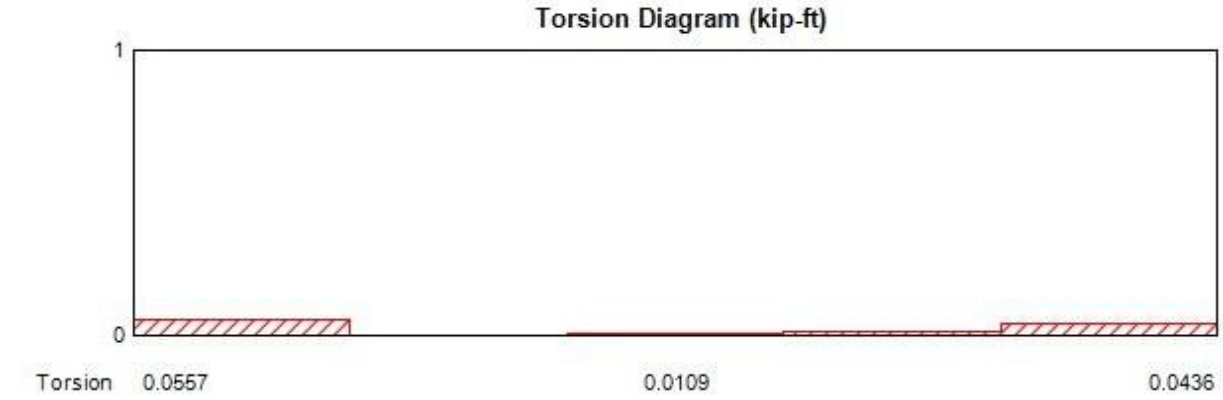
As (top)	1.9369	0.7413	1.7524
As (bot)	1.297	0.9	1.3188

Shear Diagram (kip)



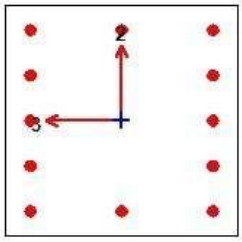
Shear	39.983	33.41	38.107


Transverse Reinforcement (in²/ft)
1
0
Asv/s reqd 0.4739 0.396 0.4516


Torsion Diagram (kip-ft)
1
0
Torsion 0.0557 0.0109 0.0436

ETABS Concrete Frame Design

ACI 318-08 Column Section Design (Summary)



Column Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (in)	LLRF	Type
4TH	C13	78	C3 24*24	1.246D-1E+1L	0	120	0.471	Sway Special

Section Properties

b (in)	h (in)	dc (in)	Cover (Torsion) (in)
24	24	2.5688	2

Material Properties

E_c (lb/in²)	f'_c (lb/in²)	Lt.Wt Factor (Unitless)	f_y (lb/in²)	f_{ys} (lb/in²)
3604997	4000	1	60000	60000

Design Code Parameters

ϕ_T	ϕ_{CTied}	$\phi_{CSpiral}$	ϕ_{Vns}	ϕ_{Vs}	ϕ_{Vjoint}
0.9	0.65	0.75	0.75	0.6	0.85

Axial Force and Biaxial Moment Design for P_u , M_{u2} , M_{u3}

Design P_u kip	Design M_{u2} kip-ft	Design M_{u3} kip-ft	Minimum M2 kip-ft	Minimum M3 kip-ft	Rebar Area in²	Rebar % %
395.567	-43.5123	135.7652	43.5123	43.5123	5.76	1.25

Axial Force and Biaxial Moment Factors

	C_m Factor Unitless	δ_{ns} Factor Unitless	δ_s Factor Unitless	K Factor Unitless	Length in
Major Bend(M3)	0.29617	1	1	1	96
Minor Bend(M2)	0.225547	1	1	1	96

Shear Design for V_{u2} , V_{u3}

	Shear V_u kip	Shear ϕV_c kip	Shear ϕV_s kip	Shear ϕV_p kip	Rebar A_v/s in²/ft
Major, V_{u2}	29.861	65.55	0	0	0
Minor, V_{u3}	27.724	64.47	0	0	0

Joint Shear Check/Design

	Joint Shear Force kip	Shear $V_{u,Top}$ kip	Shear $V_{u,Tot}$ kip	Shear ϕV_c kip	Joint Area in²	Shear Ratio Unitless
Major Shear, V_{u2}	N/N	N/N	N/N	N/N	N/N	N/N
Minor Shear, V_{u3}	N/N	N/N	N/N	N/N	N/N	N/N

(6/5) Beam/Column Capacity Ratio

Major Ratio	Minor Ratio
N/N	N/N

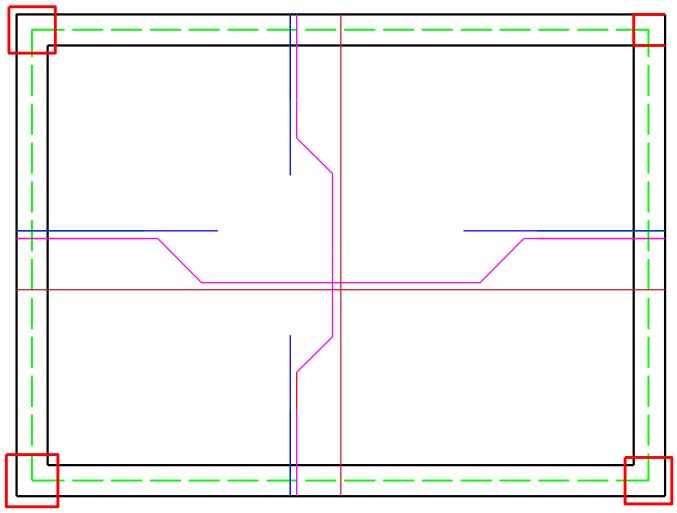
Notes:

N/A: Not Applicable

N/C: Not Calculated

N/N: Not Needed

REINFORCEMENT
DETAILING
OF
SLAB



LEGEND:

#3straight bar
@10"c/c

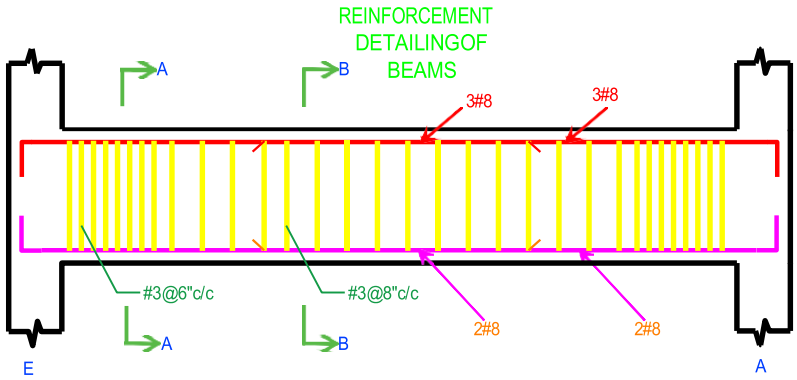
#3cranked bar

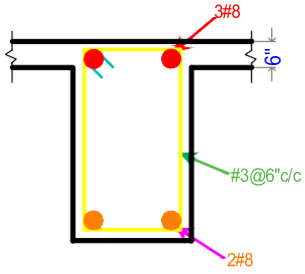
#3 extra top
@20"c/c

NORTH

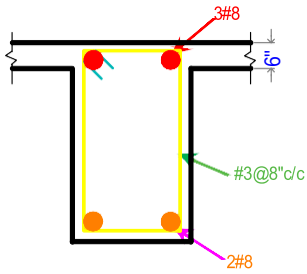
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REINFORCEMENT
DETAILINGOF
BEAMS





SECTIONA-A



SECTIONB-B

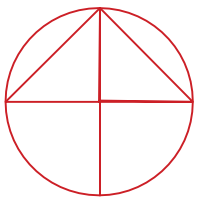
LEGEND:

Negative Longitudinal Reinforcement

Positive Longitudinal Reinforcement

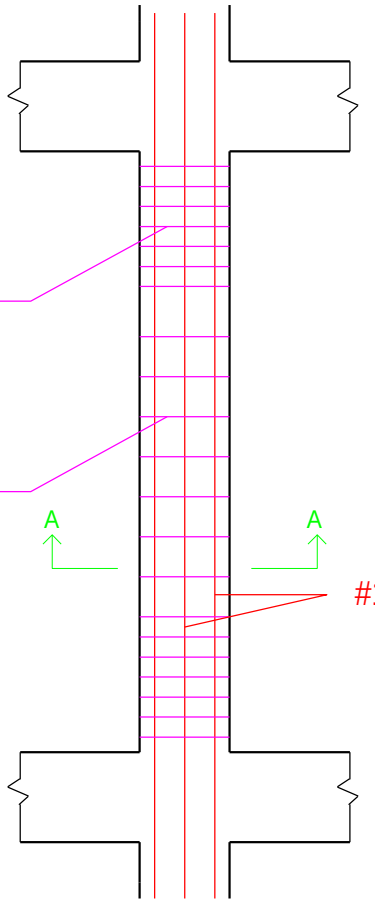
Shear Reinforcement

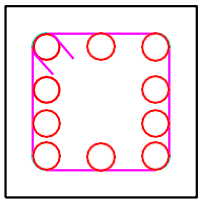
NORTH



ID:210051162

REINFORCEMENT
DETAILINGOF
COLUMN





8 #10 Bar

Section A-A

Cross-section

longitudinal section

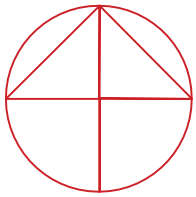
LEGEND:

Longitudinal Reinforcement

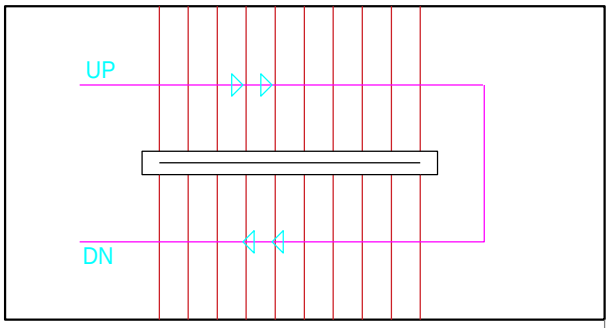
Longitudinal Reinforcement (Section view)

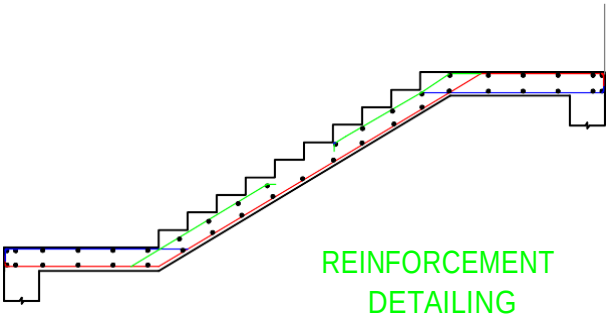
Shear Reinforcement (Stirrups)

NORTH



ID:210051162



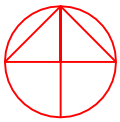


REINFORCEMENT
DETAILING
OF
STAIRS

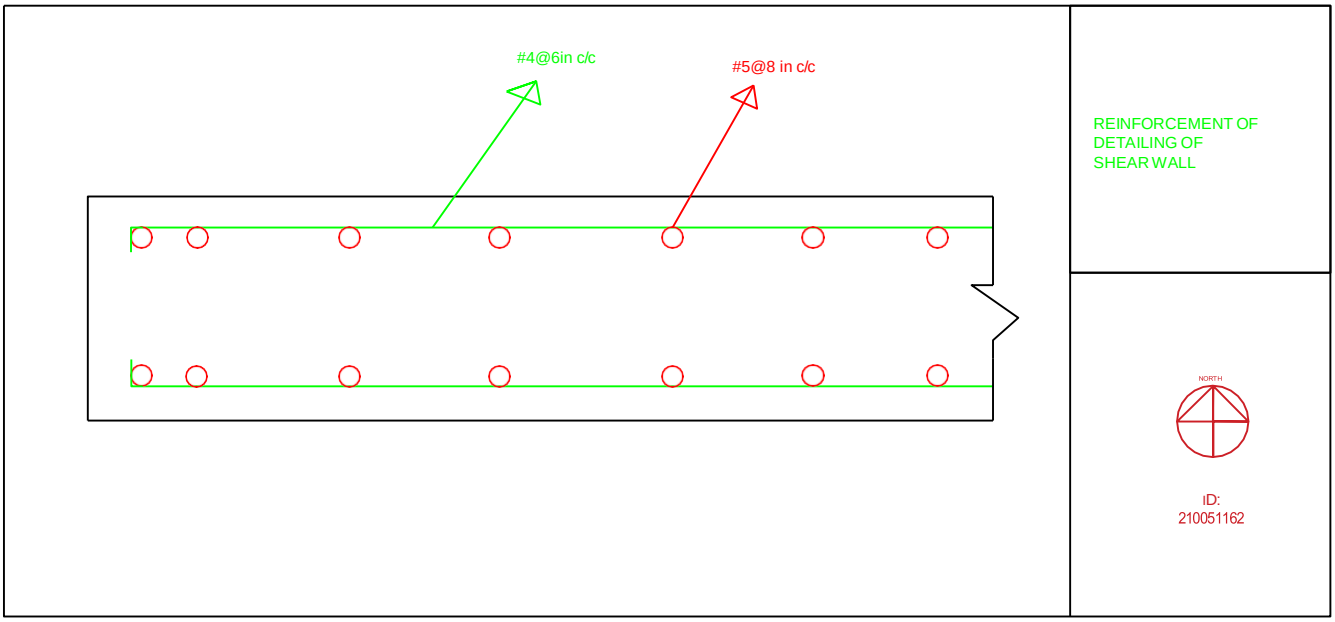
LEGEND:

- #4@10"c/c
- #3@7"c/c
- #4@10"c/c
- #3@8"c/c(stair)
- #3@7"c/c(landing)

NORTH



ID:210051162


#4@6in c/c
#5@8 in c/c
REINFORCEMENT OF
DETAILING OF
SHEAR WALL
NORTH
ID:
210051162

Discussion

Structural Analysis and Design Methodology

The structural integrity of the proposed reinforced concrete (RC) residential building was assessed using **ETABS**, with the design framework strictly adhering to the **Bangladesh National Building Code (BNBC) 2020**. To ensure accuracy, the digital twin was developed from architectural blueprints, incorporating precise modeling assumptions for material properties, cross-sectional dimensions, and boundary conditions.

The analysis evaluated the building's response to a comprehensive suite of load combinations, including gravity (dead and live) and lateral (wind and seismic) forces. Beyond strength verification, serviceability was a primary focus; parameters such as **storey drift, lateral displacement, torsional irregularity, and soft storey effects** were rigorously checked. These ETABS outputs were further validated through independent calculations in Microsoft Excel to ensure compliance with code-defined limits.

An iterative design approach was adopted. Initial simulations led to strategic adjustments in the structural layout to optimize load paths and overall stability. Once the model demonstrated satisfactory performance, all primary components—including beams, columns, slabs, shear walls, and staircases—were designed. These results were then manually audited against software outputs to confirm reliability. Finally, detailing was executed to ensure that reinforcement layouts are not only code-compliant but also highly constructible for on-site implementation.

Conclusion

Comprehensive structural evaluation and design of the proposed reinforced concrete residential building confirm full compliance with the **BNBC 2020** standards. The structure was rigorously tested against both **strength and serviceability requirements**, with all parameters—including lateral drift and vertical deflection—falling well within permissible limits. By cross-referencing **ETABS analytical data** with manual design calculations for critical members, the reliability of the structural model was successfully validated. Consequently, the proposed system is deemed safe and structurally sound for implementation, contingent upon strict adherence to the specified material grades and construction quality standards.

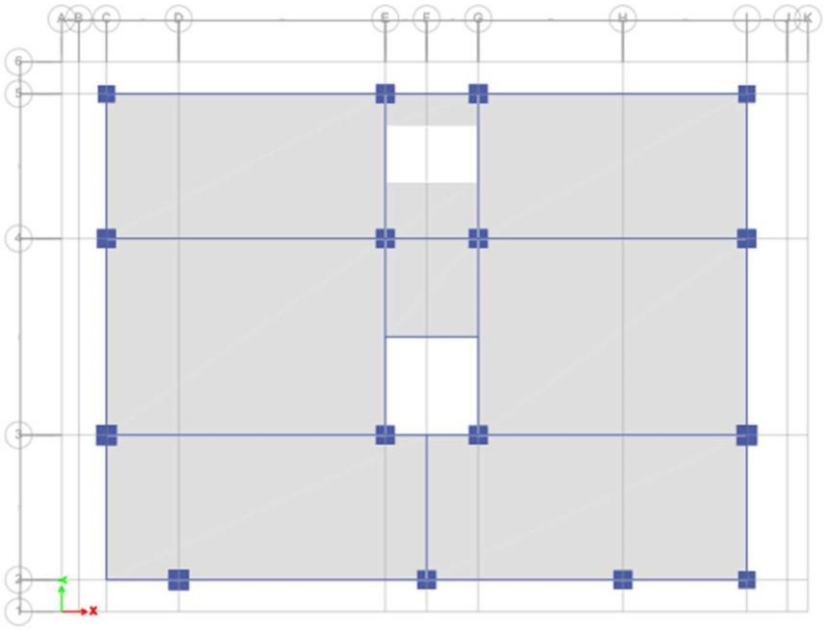
References

Title	BNBC 20 PAGE NUMBER	Reference
Domestic Purposes in Residential Buildings	4815	**Ministry of Housing and Public Works, Government of the People's Republic of Bangladesh. (2020). Bangladesh National Building Code (BNBC). Government of Bangladesh**
Main wind-force resisting system	3128	
Basic Wind Speed	3130	
Wind directionality factor	3131	
Importance Factor	3132	
Gust Effect Factor	3134	
Dead Loads	3110	
Weight of Construction Materials	3111	
Unit Weight of Basic Materials	3110	
Live Loads	3113	
Minimum Uniformly Distributed and Concentrated Live Loads	3114	
Provision for Partition Walls	3119	
Design base shear	3207	
Design response spectrum	3192	
Seismic zoning map of Bangladesh	3195	
Seismic Zone Coefficient Z for Some Important Towns of Bangladesh	3196	
Site classification	3189	
Site Classification Based on Soil Properties	3190	
Site Dependent Soil Factor and Other Parameters Defining Elastic Response Spectrum	3197	
Occupancy Category of Buildings and other Structures for Flood, Surge, Wind and Earthquake Loads.	3061	

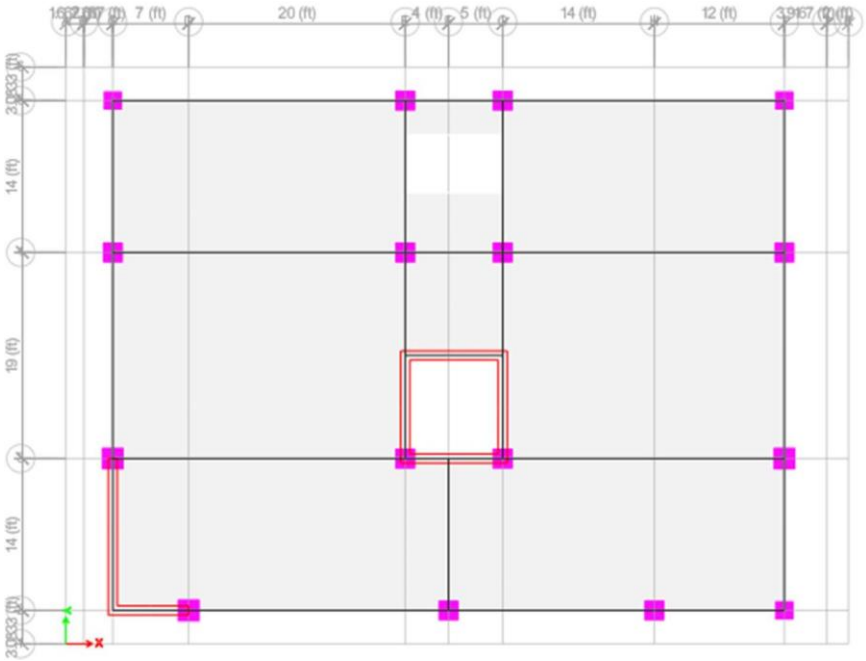
Importance Factors for Buildings and Structures for Earthquake design	3197	
Response Reduction Factor, Deflection Amplification Factor and Height Limitations for Different Structural Systems	3202	
Building period	3208	
Values for Coefficients to Estimate Approximate Period	3209	
Seismic weight		
Combinations of Loads	3264	
Symbols and Notation	3104	
Horizontal earthquake loading, Eh	3226	
Seismic Design Category of Buildings	3198	
Vertical earthquake loading, Ev	3226	
Plan (Horizontal) Irregularities of Structures	3072	
Accidental torsional effects	3211	
Design for torsional effects	3212	
Vertical distribution of lateral forces	3210	
Drift and Building Separation	3080	
Story drift limitation		
Deflection and story drift	3212	
Allowable Story Drift Limit	3229	

APPENDIX

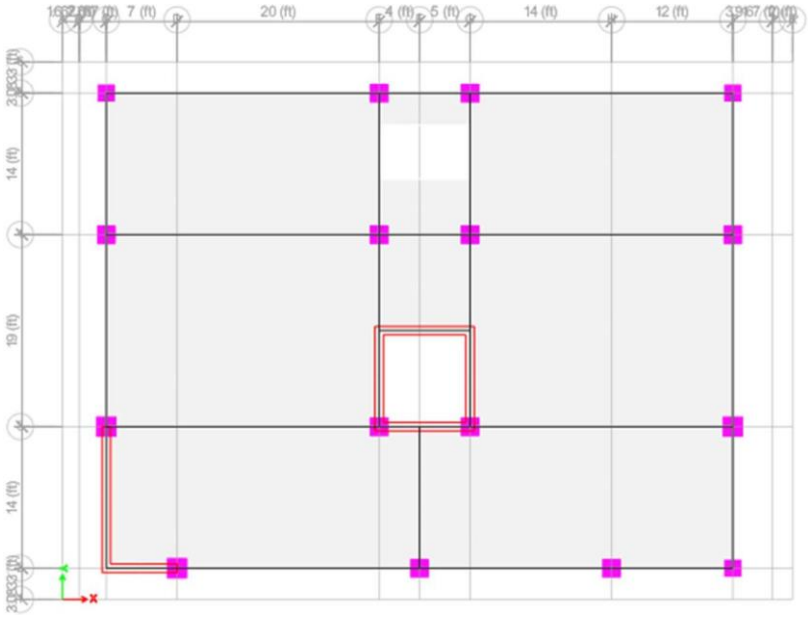
BEFORE SHEAR WALL:



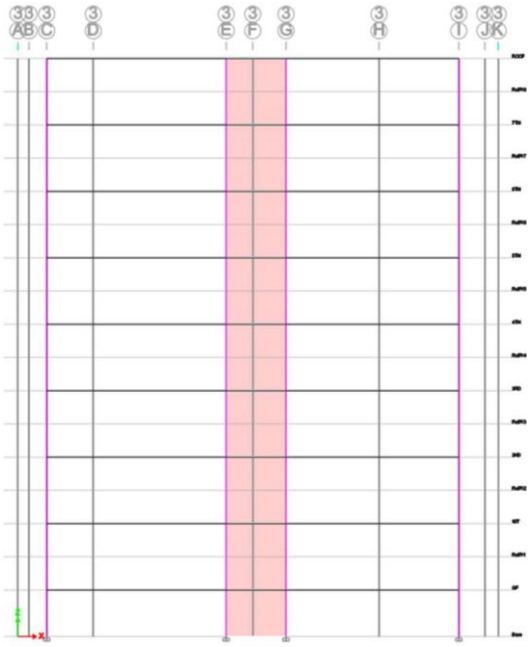
AFTER SHEAR WALL:



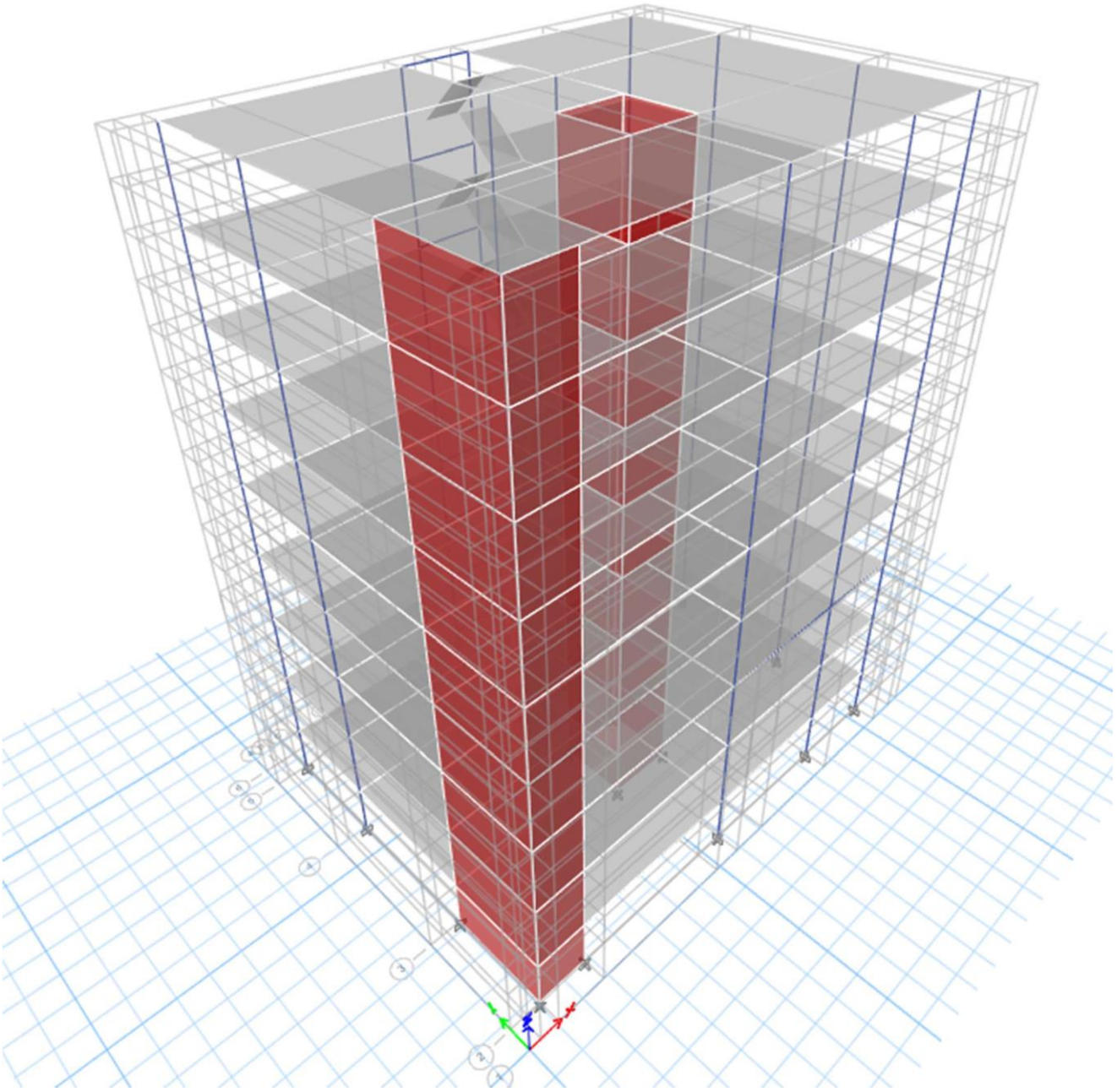
PLAN VIEW:



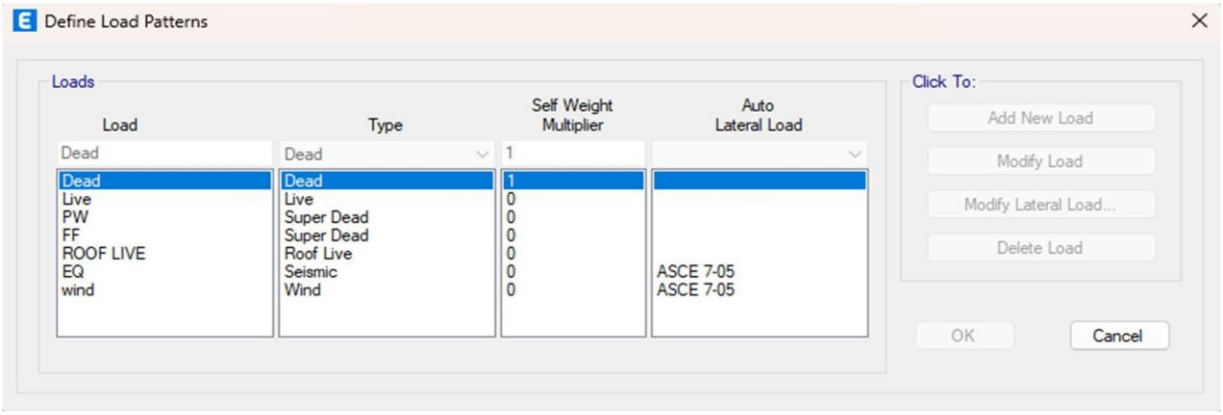
ELEVATION VIEW:



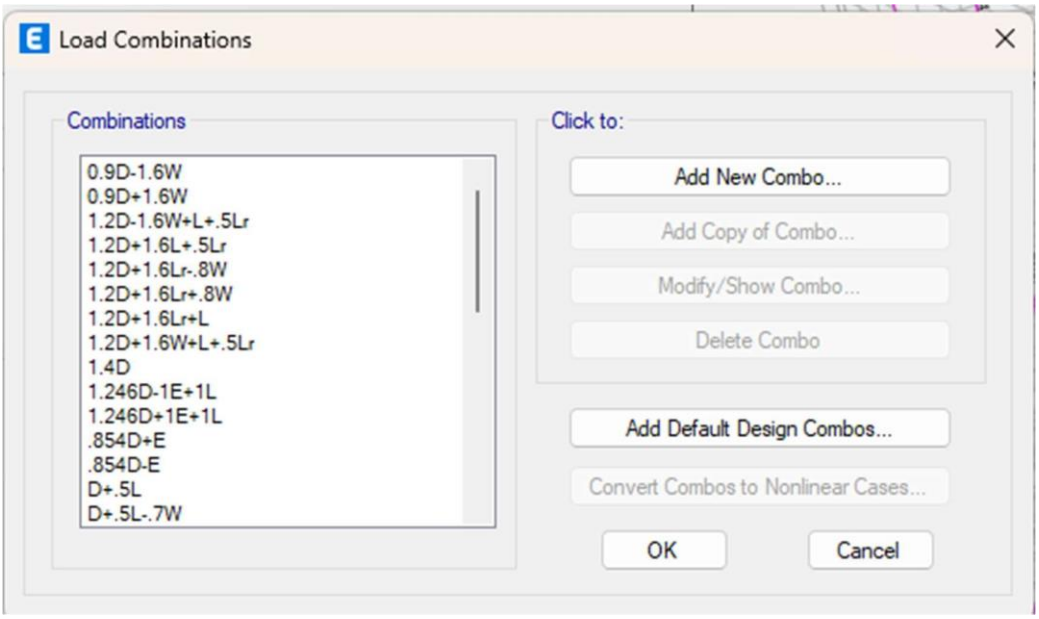
3D VIEW:



LOAD PATTERNS:

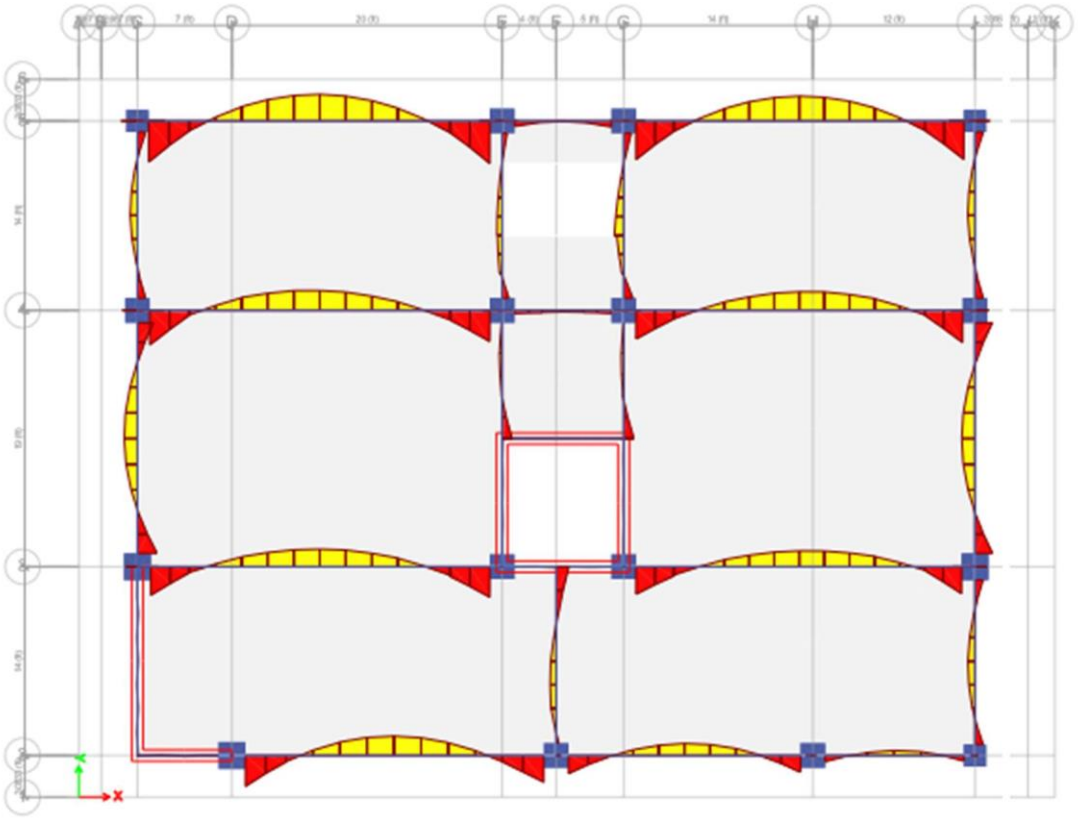


LOAD COMBINATIONS:

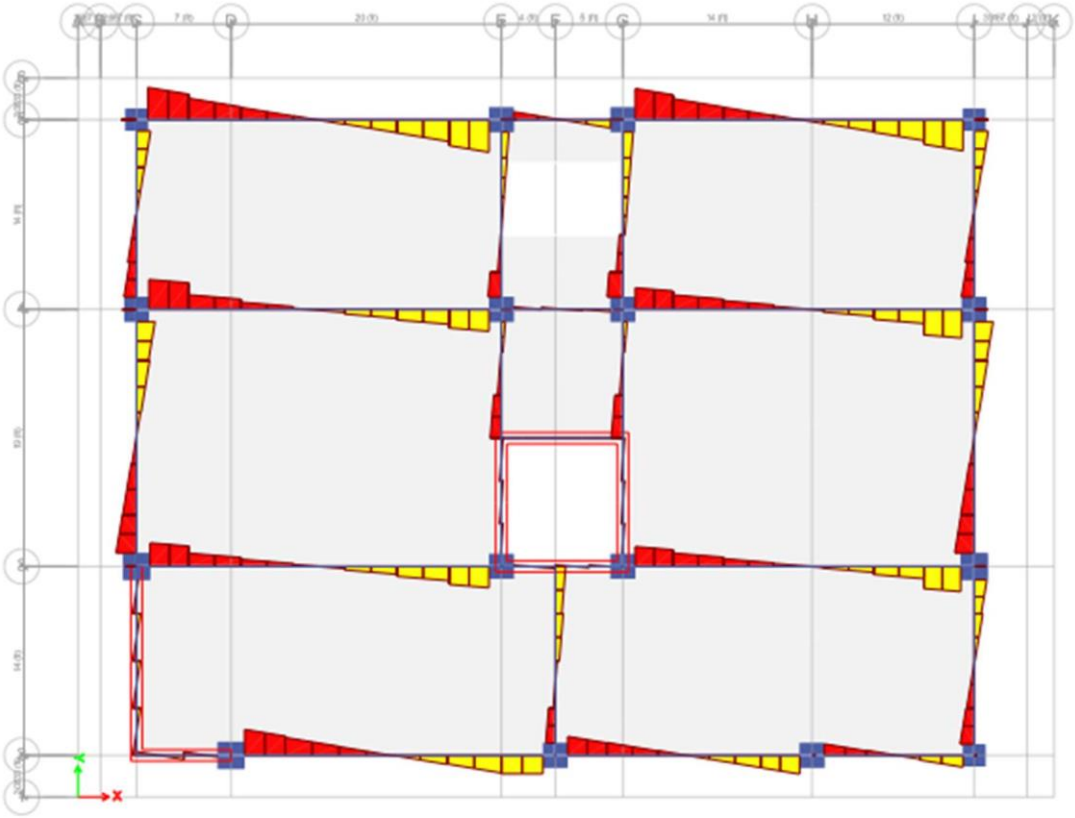


DIAGRAMS:

BENDING MOMENT DIAGRAM:



SHEAR FORCE DIAGRAM:



RESTRAINED REACTIONS (Fz):

